

Fail2ban: базовая защита сервера

Лабораторная работа №16

Новичков Максим

Российский
университет дружбы
народов

Цель работы

Получить навыки использования Fail2ban для защиты сервисов от атак типа *brute force*.

Выполнение лабораторной работы

Установка Fail2ban

```
Running transaction:
Preparing      :                               1/1
Running scriptlet: fail2ban-selinux-1.1.0-6.el10_0.noarch 1/5
Installing     : fail2ban-selinux-1.1.0-6.el10_0.noarch 1/5
Running scriptlet: fail2ban-selinux-1.1.0-6.el10_0.noarch 1/5
Installing     : fail2ban-server-1.1.0-6.el10_0.noarch 2/5
Running scriptlet: fail2ban-server-1.1.0-6.el10_0.noarch 2/5
Installing     : fail2ban-firewalld-1.1.0-6.el10_0.noarch 3/5
Installing     : fail2ban-sendmail-1.1.0-6.el10_0.noarch 4/5
Installing     : fail2ban-1.1.0-6.el10_0.noarch 5/5
Running scriptlet: fail2ban-selinux-1.1.0-6.el10_0.noarch 5/5
Running scriptlet: fail2ban-1.1.0-6.el10_0.noarch 5/5

Installed:
fail2ban-1.1.0-6.el10_0.noarch                fail2ban-firewalld-1.1.0-6.el10_0.noarch
fail2ban-selinux-1.1.0-6.el10_0.noarch        fail2ban-sendmail-1.1.0-6.el10_0.noarch
fail2ban-server-1.1.0-6.el10_0.noarch

Complete!
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# systemctl start fail2ban.service
[root@server.manovichkov.net ~]# systemctl enable fail2ban.service
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
```

Рис. 1: Установка и запуск Fail2ban

Просмотр журнала Fail2ban

```
[manovichkov@server.manovichkov.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for manovichkov:
2025-11-09 10:29:45,874 fail2ban.server      [16155]: INFO    -----
----
2025-11-09 10:29:45,875 fail2ban.server      [16155]: INFO    Starting Fail2ban v1.1.0
2025-11-09 10:29:45,875 fail2ban.observer    [16155]: INFO    Observer start...
2025-11-09 10:29:45,878 fail2ban.database    [16155]: INFO    Connected to fail2ban persistent database '/va
r/lib/fail2ban/fail2ban.sqlite3'
2025-11-09 10:29:45,879 fail2ban.database    [16155]: WARNING New database created. Version '4'
```

Рис. 2: Первичный журнал Fail2ban

Создание локальной конфигурации

```
customisation.local  [----] 14 L:[ 1+12 13/ 13] *(135 / 135b) <EOF> [*][X]  
[DEFAULT]  
bantime = 3600  
  
[sshd]  
port = ssh,2022  
enabled = true  
  
[sshd-ddos]  
filter = sshd  
enabled = true  
  
[selinux-ssh]  
enabled = true
```

Запуск jail'ов SSH

```
2025-11-09 10:33:44,197 fail2ban.jail [17151]: INFO Creating new jail 'sshd'
2025-11-09 10:33:44,199 fail2ban.jail [17151]: INFO Jail 'sshd' uses systemd {}
2025-11-09 10:33:44,199 fail2ban.jail [17151]: INFO Initiated 'systemd' backend
2025-11-09 10:33:44,200 fail2ban.filter [17151]: INFO maxLines: 1
2025-11-09 10:33:44,204 fail2ban.filtersystemd [17151]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session'
2025-11-09 10:33:44,204 fail2ban.filter [17151]: INFO maxRetry: 5
2025-11-09 10:33:44,204 fail2ban.filter [17151]: INFO findtime: 600
2025-11-09 10:33:44,204 fail2ban.actions [17151]: INFO banTime: 3600
2025-11-09 10:33:44,204 fail2ban.filter [17151]: INFO encoding: UTF-8
2025-11-09 10:33:44,205 fail2ban.jail [17151]: INFO Creating new jail 'selinux-ssh'
2025-11-09 10:33:44,207 fail2ban.jail [17151]: INFO Jail 'selinux-ssh' uses pyinotify {}
2025-11-09 10:33:44,208 fail2ban.jail [17151]: INFO Initiated 'pyinotify' backend
2025-11-09 10:33:44,208 fail2ban.datedetector [17151]: INFO date pattern '%%': 'Epoch'
2025-11-09 10:33:44,208 fail2ban.filter [17151]: INFO maxRetry: 5
2025-11-09 10:33:44,208 fail2ban.filter [17151]: INFO findtime: 600
2025-11-09 10:33:44,208 fail2ban.actions [17151]: INFO banTime: 3600
2025-11-09 10:33:44,208 fail2ban.filter [17151]: INFO encoding: UTF-8
2025-11-09 10:33:44,209 fail2ban.filter [17151]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 0, hash = 6b521be27cca9b97d1007e5f308bb207d42e39b0)
2025-11-09 10:33:44,210 fail2ban.jail [17151]: INFO Creating new jail 'sshd-ddos'
2025-11-09 10:33:44,210 fail2ban.jail [17151]: INFO Jail 'sshd-ddos' uses pyinotify {}
2025-11-09 10:33:44,210 fail2ban.jail [17151]: INFO Initiated 'pyinotify' backend
2025-11-09 10:33:44,211 fail2ban.filter [17151]: INFO maxLines: 1
2025-11-09 10:33:44,211 fail2ban.filter [17151]: INFO maxRetry: 5
2025-11-09 10:33:44,211 fail2ban.filter [17151]: INFO findtime: 600
2025-11-09 10:33:44,211 fail2ban.actions [17151]: INFO banTime: 3600
2025-11-09 10:33:44,211 fail2ban.filter [17151]: INFO encoding: UTF-8
2025-11-09 10:33:44,212 fail2ban.jail [17151]: INFO Jail 'sshd' started
2025-11-09 10:33:44,212 fail2ban.jail [17151]: INFO Jail 'selinux-ssh' started
2025-11-09 10:33:44,213 fail2ban.jail [17151]: INFO Jail 'sshd-ddos' started
2025-11-09 10:33:44,219 fail2ban.filtersystemd [17151]: INFO [sshd] Jail is in operation now (process new journal entries)
```

Рис. 4: Запуск jail'ов SSH

Включение HTTP jail'ов

```
customisation.local [BM--] 0 L:[ 22+25 47/ 57] *(494 / 532b) 0010 0x00A
enabled = true

[apache-overflows]
enabled = true

[apache-nohome]
enabled = true

[apache-fakegooglebot]
enabled = true

[apache-modsecurity]
enabled = true

[apache-shellshock]
enabled = true

[postfix]
enabled = true

[postfix-rbl]
enabled = true

[dovecot]
enabled = true

[postfix-sasl]
enabled = true
```

Запуск HTTP- защиты

```
2025-11-09 10:37:31,869 fail2ban.jail [17732]: INFO Creating new jail 'sshd-ddos'
2025-11-09 10:37:31,869 fail2ban.jail [17732]: INFO Jail 'sshd-ddos' uses pyinotify {}
2025-11-09 10:37:31,869 fail2ban.jail [17732]: INFO Initiated 'pyinotify' backend
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO maxLines: 1
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO maxRetry: 5
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO findtime: 600
2025-11-09 10:37:31,870 fail2ban.actions [17732]: INFO banTime: 3600
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO encoding: UTF-8
2025-11-09 10:37:31,871 fail2ban.filtersystemd [17732]: INFO [sshd] Jail is in operation now (process new j
ournal entries)
2025-11-09 10:37:31,871 fail2ban.jail [17732]: INFO Jail 'sshd' started
2025-11-09 10:37:31,871 fail2ban.jail [17732]: INFO Jail 'selinux-ssh' started
2025-11-09 10:37:31,872 fail2ban.jail [17732]: INFO Jail 'apache-auth' started
2025-11-09 10:37:31,872 fail2ban.jail [17732]: INFO Jail 'apache-badbots' started
2025-11-09 10:37:31,873 fail2ban.jail [17732]: INFO Jail 'apache-noscript' started
2025-11-09 10:37:31,873 fail2ban.jail [17732]: INFO Jail 'apache-overflows' started
2025-11-09 10:37:31,873 fail2ban.jail [17732]: INFO Jail 'apache-nohome' started
2025-11-09 10:37:31,874 fail2ban.jail [17732]: INFO Jail 'apache-fakegooglebot' started
2025-11-09 10:37:31,875 fail2ban.jail [17732]: INFO Jail 'apache-modsecurity' started
2025-11-09 10:37:31,875 fail2ban.jail [17732]: INFO Jail 'apache-shellshock' started
2025-11-09 10:37:31,876 fail2ban.jail [17732]: INFO Jail 'postfix' started
2025-11-09 10:37:31,876 fail2ban.jail [17732]: INFO Jail 'postfix-rbl' started
2025-11-09 10:37:31,877 fail2ban.filtersystemd [17732]: INFO [postfix] Jail is in operation now (process ne
w journal entries)
2025-11-09 10:37:31,877 fail2ban.filtersystemd [17732]: INFO [postfix-rbl] Jail is in operation now (proces
s new journal entries)
2025-11-09 10:37:31,877 fail2ban.jail [17732]: INFO Jail 'dovecot' started
2025-11-09 10:37:31,877 fail2ban.filtersystemd [17732]: INFO [postfix-sasl] Jail is in operation now (proces
s new journal entries)
2025-11-09 10:37:31,877 fail2ban.filtersystemd [17732]: INFO [dovecot] Jail is in operation now (process ne
w journal entries)
2025-11-09 10:37:31,878 fail2ban.jail [17732]: INFO Jail 'postfix-sasl' started
2025-11-09 10:37:31,878 fail2ban.jail [17732]: INFO Jail 'sshd-ddos' started
```

Рис. 6: Запуск HTTP-jail'ов

Активация Postfix и Dovecot

```
customisation.local [BM--] 0 L:[ 22+25 47/ 57] *(494 / 532b) 0010 0x00A
enabled = true

[apache-overflows]
enabled = true

[apache-nohome]
enabled = true

[apache-fakegooglebot]
enabled = true

[apache-modsecurity]
enabled = true

[apache-shellshock]
enabled = true

[postfix]
enabled = true

[postfix-irbl]
enabled = true

[dovecot]
enabled = true

[postfix-sasl]
enabled = true
```

Рис. 7: Запуск jail'ов для почтовых сервисов

Журнал после настройки почты

```
2025-11-09 10:37:31,869 fail2ban.jail [17732]: INFO Creating new jail 'sshd-ddos'
2025-11-09 10:37:31,869 fail2ban.jail [17732]: INFO Jail 'sshd-ddos' uses pyinotify {}
2025-11-09 10:37:31,869 fail2ban.jail [17732]: INFO Initiated 'pyinotify' backend
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO maxLines: 1
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO maxRetry: 5
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO findtime: 600
2025-11-09 10:37:31,870 fail2ban.actions [17732]: INFO banTime: 3600
2025-11-09 10:37:31,870 fail2ban.filter [17732]: INFO encoding: UTF-8
2025-11-09 10:37:31,871 fail2ban.filterssystemd [17732]: INFO [sshd] Jail is in operation now (process new j
ournal entries)
2025-11-09 10:37:31,871 fail2ban.jail [17732]: INFO Jail 'sshd' started
2025-11-09 10:37:31,871 fail2ban.jail [17732]: INFO Jail 'selinux-ssh' started
2025-11-09 10:37:31,872 fail2ban.jail [17732]: INFO Jail 'apache-auth' started
2025-11-09 10:37:31,872 fail2ban.jail [17732]: INFO Jail 'apache-badbots' started
2025-11-09 10:37:31,873 fail2ban.jail [17732]: INFO Jail 'apache-noscript' started
2025-11-09 10:37:31,873 fail2ban.jail [17732]: INFO Jail 'apache-overflows' started
2025-11-09 10:37:31,873 fail2ban.jail [17732]: INFO Jail 'apache-nohome' started
2025-11-09 10:37:31,874 fail2ban.jail [17732]: INFO Jail 'apache-fakegooglebot' started
2025-11-09 10:37:31,875 fail2ban.jail [17732]: INFO Jail 'apache-modsecurity' started
2025-11-09 10:37:31,875 fail2ban.jail [17732]: INFO Jail 'apache-shellshock' started
2025-11-09 10:37:31,876 fail2ban.jail [17732]: INFO Jail 'postfix' started
2025-11-09 10:37:31,876 fail2ban.jail [17732]: INFO Jail 'postfix-rbl' started
2025-11-09 10:37:31,877 fail2ban.filterssystemd [17732]: INFO [postfix] Jail is in operation now (process ne
w journal entries)
2025-11-09 10:37:31,877 fail2ban.filterssystemd [17732]: INFO [postfix-rbl] Jail is in operation now (proces
s new journal entries)
2025-11-09 10:37:31,877 fail2ban.jail [17732]: INFO Jail 'dovecot' started
2025-11-09 10:37:31,877 fail2ban.filterssystemd [17732]: INFO [postfix-sasl] Jail is in operation now (proce
ss new journal entries)
2025-11-09 10:37:31,877 fail2ban.filterssystemd [17732]: INFO [dovecot] Jail is in operation now (process ne
w journal entries)
2025-11-09 10:37:31,878 fail2ban.jail [17732]: INFO Jail 'postfix-sasl' started
2025-11-09 10:37:31,878 fail2ban.jail [17732]: INFO Jail 'sshd-ddos' started
```

Рис. 8: Журнал Fail2ban после настройки почты

Общий статус Fail2ban

```
[root@server.manovichkov.net ~]# fail2ban-client status
Status
|- Number of jail:      15
`- Jail list:  apache-auth, apache-badbots, apache-fakegooglebot, apache-modsecurity, apache-nohome, apache-no
script, apache-overflows, apache-shellshock, dovecot, postfix, postfix-rbl, postfix-sasl, selinux-ssh, sshd, ss
hd-ddos
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed:     0
| `-- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
  |- Currently banned: 0
  |- Total banned:     0
  `-- Banned IP list:
[root@server.manovichkov.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 1
| |- Total failed:     3
| `-- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
  |- Currently banned: 1
  |- Total banned:     1
  `-- Banned IP list:  192.168.1.31
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# fail2ban-client set sshd unba
```

Рис. 9: Статус Fail2ban

Статус защиты SSH

```
customisation.local [-M--] 32 L:[ 1+ 3 4/ 59] *(58 / 566b) 0010 0x00A  
[DEFAULT]  
bantime = 3600  
  
ignoreip = 127.0.0.1/8 192.168.1  
  
[sshd]  
port = ssh,2022  
enabled = true  
  
[sshd-ddos]  
filter = sshd  
enabled = true
```

Рис. 10: SSH до изменения

```
[root@server.manovichkov.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed:    0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:    0
   `-- Banned IP list:
```

Рис. 11: Блокировка
IP

Запись о блокировке

```
customisation.local [-M--] 32 L:[ 1+ 3 4/ 59] *(58 / 566b) 0010 0x00A
```

```
[DEFAULT]
```

```
bantime = 3600
```

```
ignoreip = 127.0.0.1/8 192.168.1
```

```
[sshd]
```

```
port = ssh,2022
```

```
enabled = true
```

```
[sshd-ddos]
```

```
filter = sshd
```

```
enabled = true
```


Разблокировка адреса

```
2025-11-09 10:41:09,273 fail2ban.jail      [18273]: INFO    Jail 'postfix-sasl' started
2025-11-09 10:41:09,273 fail2ban.jail      [18273]: INFO    Jail 'sshd-ddos' started
2025-11-09 10:41:18,401 fail2ban.filter  [18273]: INFO    [sshd] Ignore 192.168.1.31 by ip
2025-11-09 10:41:22,384 fail2ban.filter  [18273]: INFO    [sshd] Ignore 192.168.1.31 by ip
2025-11-09 10:41:26,623 fail2ban.filter  [18273]: INFO    [sshd] Ignore 192.168.1.31 by ip
```

Рис. 13: Разблокировка

Добавление ignoreip

```
[root@server.manovichkov.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:    0
|  `- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:    0
   `- Banned IP list:
```

Рис. 14: ignoreip

Проверка статуса после изменения

```
[root@server.manovichkov.net ~]# cd /vagrant/provision/server/  
[root@server.manovichkov.net server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d  
[root@server.manovichkov.net server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provision/server/  
protect/etc/fail2ban/jail.d/  
[root@server.manovichkov.net server]# touch protect.sh  
[root@server.manovichkov.net server]#
```

Рис. 15: Статус после ignoreip

Вывод

- Установлен и настроен Fail2ban.
- Активирована защита SSH, HTTP и почтовых сервисов.
- Проверена работа блокировок и исключений IP.
- Настройки сохранены во внутреннем окружении виртуальной машины.

Fail2ban корректно обнаруживает и блокирует подозрительную активность, обеспечивая базовую защиту сервера.